[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

SHEARING MACHINES, METAL, CIRCULAR, POWER-DRIVEN

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers power-driven, circular, metal-shearing machines used for ring and circle shearing, flange-forming, and joggling operations.

2. CLASSIFICATION. The circular shearing machines shall be of the following types and sizes. The sizes shall be as specified in tables I, II, and III. Size refers to the metal thickness and minimum and maximum circle-cutting diameters the machine can accommodate. The type and size to be furnished shall be as specified (see 7.2(b)).

Type I - Shearing
Type II - Shearing and flanging
Type III - Shearing, flanging, and joggling

Size – (See tables I through III)

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3445

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TABLE I. Type I machine sizes.

Characteristics	Sizes				
	1	2	3	4	5
Workpiece thickness, inch	3/16	3/16	3/16	3/8	3/8
Circle diameter, inches:					
Minimum	12	12	12	15	15
Maximum	39	51	62	100	144
Throat depth, inches, minimum	10	23	12	12	23
Motor - horsepower, minimum	1/3	3/4	1 1/2	10	10

TABLE II. Type II machine sizes.

Characteristics	Sizes				
	1	2	3	4	5
Workpiece thickness, inch	3/16	3/16	3/16	3/8	3/8
Circle diameter, inches:					
Minimum	12	12	12	15	15
Maximum	39	51	62	100	144
Throat depth, inches, minimum	10	23	12	12	23
Flange depth, inch, minimum	1/4	1/2	1/2	1	1
Motor - horsepower, minimum	1/3	3/4	1 1/2	10	10

TABLE III. Type III machine sizes.

Characteristics	Size					
	1	2	3	4	5	6
Workpiece thickness, inch	1/8	3/16	3/8	1/2	3/4	1
Circle diameter, inches:						
Minimum	12	12	12	20	24	26
Maximum	62	80	96	144	144	144
Throat depth, inches, minimum	20	36	36	48	48	48
Flange depth, inches, minimum	1	1	1/4	2	2	2
Joggling thickness, inch, minimum	12 ga.	1/8	1/4	3/8	1/2	5/8
Straight cut width range, inches	1 – 26	1 1/2 – 26	2 1/2 – 40	2 1/2 – 40	3 – 40	5 – 40
Motor - horsepower, minimum	1/3	3/4	1 1/2	10	10	10

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3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. The machine sizes shall conform to tables I through III. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing.

3.2 Components. The shearing machine shall include, as a minimum, the following components:

3.2.1 Frame. The frame shall have provisions for leveling and anchoring the machine to the floor or a foundation.

3.2.2 Worksheet support. The worksheet support shall have a clamp for securing the center of the worksheet to the support and shall allow worksheet rotation. The worksheet support shall be horizontally adjustable to allow for varying diameters of cuts on machined bearing surfaces. When specified (see 7.2(c)), the clamp shall be power driven.

3.2.3 Cutter heads. The cutter heads shall be easily adaptable for the attachment of shear blades and forming (flanging and joggling) attachments, as applicable for the machine type specified. Either the upper, lower, or both cutter heads shall be power driven. Means shall be provided for laterally adjusting either cutter or both to maintain alignment between the shear blades or the forming attachments. The upper head shall be vertically adjustable and shall be provided with a means for locking the cutter head into a selected position. The upper cutter head on machine sizes with shearing capacities greater than 1/4 inch shall be provided with a power-driven down-feed mechanism for feeding the shear cutters or forming attachments as the cut or flange progresses.

3.2.4 Shear blades. The shearing machine shall be provided with upper and lower shear blades as normally furnished by the manufacturer. If required, shear blades shall be provided as specified (see 7.2(d)).

3.2.5 Flanging rolls and attachments. Type II and III circular shearing machines shall be provided with flanging roller attachments to meet the flange requirements shown in tables II and III.

3.2.6 Joggling rolls and attachments. Type III circular shearing machines shall be provided with joggling attachments that meet the joggling requirements shown in table III.

3.2.7 Operator controls. The shearing machine controls shall be convenient to the operator. Unless otherwise specified (see 7.2(e)), the manufacturer's standard control system shall consist of push buttons and selector switches for start/stop and cutting speed (if variable).

3.2.8 Hydraulic system. The hydraulic system shall provide overpressure protection. A filter system shall ensure delivery of clean fluid to the hydraulic system. The hydraulic system reservoir shall have a sight gauge to indicate fluid level. A means shall be provided to drain and clean the hydraulic reservoir.

3.2.9 Electrical system. Unless otherwise specified (see 7.2 (f)), the electrical system shall conform to the American National Standards Institute/National Fire Protection Association (ANSI/NFPA) 79. The

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machine shall have a dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2 (g)). The control system shall operate on a 120-volt, single-phase, 60-hertz circuit provided from the machine main power supply.

3.2.10 Electric motors. The shearing machine's drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1. Unless otherwise specified (see 7.2(h)), the motors shall have drip-proof enclosures.

3.2.11 Optional components. Optional components listed below shall be provided as specified (see 7.2 (i)).

a. Straight cutting attachments.
b. Additional components as specified.

3.3 Performance. Unless otherwise specified (see 7.2(j)), the circular shearing machines shall meet the requirements shown in tables I through III when tested in accordance with sections 5.3.1 through 5.3.2.

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(k)).

3.5 Safety and health requirements. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Association (OSHA) 29 CFR PART 1910.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(l)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(m)), measuring and indicating devices shall be graduated in both the US and SI systems of measurement.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(n)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Machine model designation.
d. Manufacturer's serial number.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
g. National stock number.
h. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(o)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

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a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description, conform to the manufacturer's drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The circular shearing machine shall be inspected for compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(p)), the circular shearing machine shall be tested in accordance with sections 5.3.1 through 5.3.2.

5.3.1 Operational test. The shearing machine shall be operated in accordance with the manufacturer's standard operating procedures and normal environmental conditions for warm-up and run-off checks. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls and features shall be verified during the trial period.

5.3.2 Performance tests. The shearing machine shall be capable of producing circular cuts and rings without ripping or wrinkling the worksheet at the maximum rated capacity of the machine. Type II and III machines shall produce flanges in mild steel that are uniform to within 1/16 inch throughout the circumference of the circle. The type III machine shall meet the joggling requirements shown in table III.

5.4 Acceptance criteria. Unless otherwise specified (see 7.2(q)), preliminary and final acceptance testing shall be conducted at the manufacturer's site. Failure of the machine to meet the testing requirements stated in section 5 shall be cause for rejection.

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6. PACKAGING. Preservation, packing, and marking shall be as specified (see 7.2(r)).

7. NOTES

7.1 Sources of documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

National Electrical Manufacturers Association (NEMA)

NEMA MG-1 Motors and Generators

Applications for copies should be sent to National Electrical Manufacturer's Association, 2101 L Street, Washington, DC 20037.

National Fire Protection Association (NFPA)

ANSI/NFPA 79 Electrical Standard for Industrial Machinery

Applications for copies should be sent to American National Standards Institute, 11 W 42nd Street, New York, NY 10036.

Occupational Safety and Health Association (OSHA)

OSHA 1910.212 Occupational Safety and Health Standards for General Industry (29 CFR Part 1910)

Applications for copies should be sent to U.S. Labor Department, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number and date of this document.
b. Type and size of circular shearing machine required (see 2).
c. Power driven worksheet clamp, if required (see 3.2.2).
d. Shear blades, as specified (see 3.2.4).
e. Operator controls, if different (see 3.2.7).

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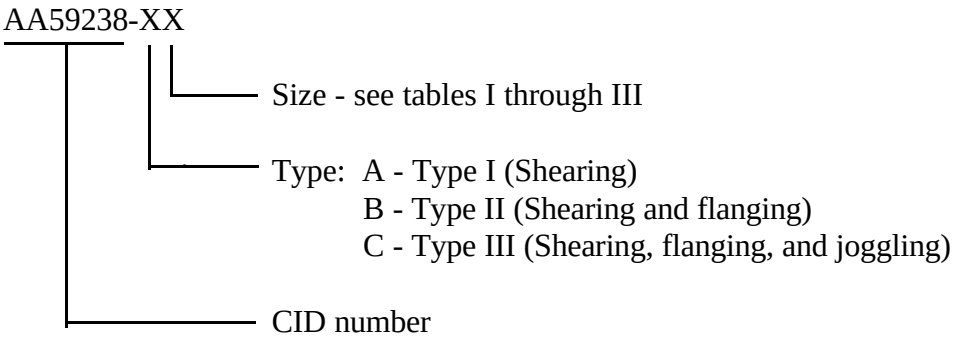
f. Electrical requirements, if different (see 3.2.9).
g. Initial voltage wiring requirements, specify (see 3.2.9)
h. Motor enclosure, if different (see 3.2.10).
i. Optional components, as specified (see 3.2.11).
j. Performance, if different (see 3.3).
k. Machine dimensions and weight, if required (see 3.4).
l. System of unit required (see 3.6).
m. Dual system of units, if required (see 3.6).
n. Nameplate, if different (see 3.7).
o. Lubrication plate or chart, if required (see 3.8).
p. Testing, if different (see 5.3).
q. Acceptance, if different (see 5.4).
r. Packaging requirements, as specified (see 6).

7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



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MILITARY INTERESTS:

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Navy - MC

PREPARING ACTIVITY:
DLA - GS

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